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Modeling of a space experiment on Bridgman solidification of concentrated semiconductor alloy

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Abstract

A GaInSb concentrated alloy has been solidified under microgravity conditions by using the Bridgman method during the STS 78/LMS mission. A numerical simulation is used in order to explain the behavior of the axial In solute distribution as revealed by analysis of the flight sample. Measurements of the gravity level onboard the Shuttle, show a significant increase of the *g*-jitter amplitude, 90 min after the start of the growth process, which is correlated with perturbation of the chemical segregation.

A numerical analysis of the *g*-jitter effect on the chemical segregation in the case of concentrated GaInSb alloy solidification is performed by varying the amplitude and the frequency of the residual gravity. A small effect of periodic accelerations on the solute distribution is observed only for frequencies below 10^{-2} Hz. At higher frequencies, the axial and radial segregation are practically unchanged.

The numerical modeling shows that the sharp increase of *g*-jitters accelerations 90 min after the start of the growth process, produces an increase of the convective intensity, which mixes the melt and increases the solidification temperature so that the interface jumps forward. The associated morphological destabilization of the interface is experimentally observed on the etched sample and on the longitudinal profile of solute concentration measured in the crystal. After a chemical homogenization of the melt, the controlled growth is started again with an axial concentration profile, corresponding to a Scheil's law.
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1. Introduction

The space experiment ESA8 performed during the STS 78/LMS mission was devoted to the study

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of Bridgman solidification of concentrated $Ga_{1-x}In_xSb$ alloys. The purpose of the experiment is to study the thermal, chemical and dewetting phenomena during Bridgman growth of concentrated alloys under microgravity conditions [1,2] as compared with similar ground experiments [2,3]. The advanced gradient heating facility (AGHF) Bridgman furnace has a heating element and a cooling zone separated by an adiabatic zone. It is pulled with a constant velocity along a tantalum cartridge containing the sample–crucible system. Two crucibles made of boron nitride (BN) and silica (SiO_2) have been used in order to act on the interface curvature during the solidification (see Fig. 1a). The GaInSb sample, 152 mm in length and 12 mm in diameter, had an initial InSb composition, measured before the experiment at the sample–seed junction: $C_0 = 19\%$. In order to avoid the undercooling phenomenon, a polycrystalline seed 63 mm in length and initial composition $C_0 = 8\%$, is used. Eight millimeter of seed are melted and mixed with the feed material before the controlled growth process to be started. Temperature measurements on the crucible external walls, have been performed during the growth process, by using a set of 9 thermocouples.

During the experiment, 3D accelerometers measured the gravity vector oscillations, however; due to technical constraints, these measurements were possible only for frequencies higher than 10^{-3} Hz [4]. For lower frequencies, the 3D gravity level was computed versus time by dedicated software NASA–PIMS team. The quasi-steady value was found to be lower than $10^{-6}g_0$ with a frequency lower than 10^{-4} Hz and no preferential orientation. From these measurements and calculations, it appeared that 90 min after the beginning

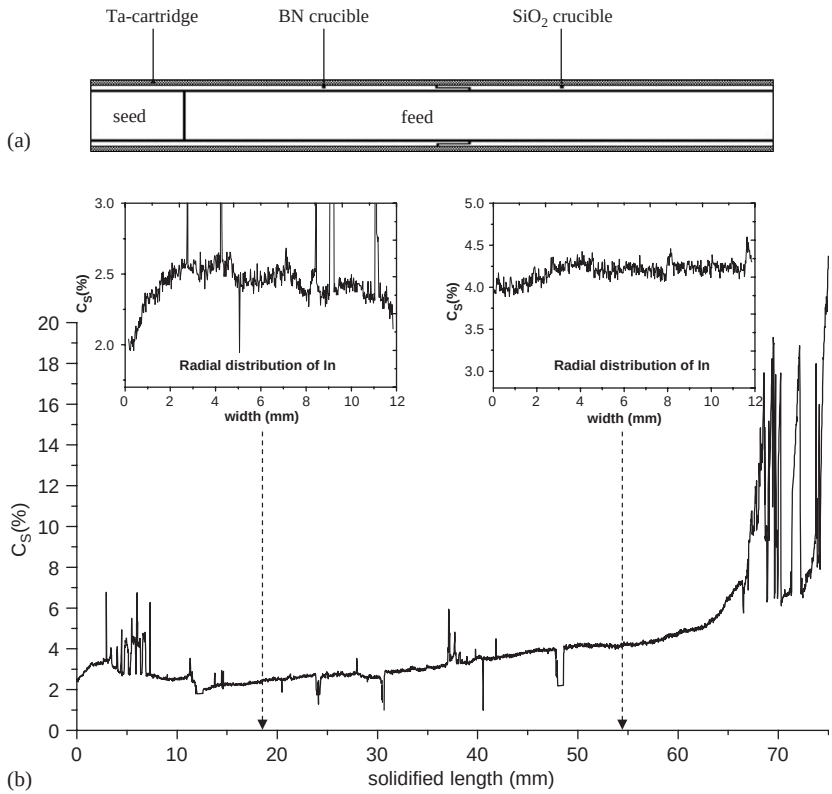

Fig. 1. (a) Schema of tantalum cartridge containing the sample-crucible system; (b) axial and radial distribution of the indium in the flight sample.

of the growth process, a 10-time increase of the acceleration power spectral density occurred.

After the experiment, the Ga, In and Sb composition of the sample is measured longitudinally and radially by the electron microprobe method. Longitudinal measurements of indium concentration at the sample center and sides show almost the same profile [2]. The axial distribution of indium in the sample shows a complicated behavior at the beginning of the solidification, corresponding to a change of the gravity level as measured in the Shuttle. Then a typical Scheil's axial profile of solute is observed, which is surprising at the low level of the measured residual gravity (see Fig. 1b). The radial variation of In concentration in the space crystal is smaller compared to the ground samples with 10% and 20% indium, but not negligible. In order to explain the segregation phenomena observed during the space experiment, a transient simulation of heat, momentum and solute transport is performed.

The analysis of steady gravity effect on the melt convection and solute segregation during $Ga_{1-x}In_xSb$ ($x = 0.1$) concentrated alloy solidification has been performed in Ref. [5] and focused on the solutal effect on the melt convection and interface curvature for a 2D horizontal Bridgman configuration at steady gravity levels between 10^{-6} and $10^{-5}g_0$. The cylindrical sample is modeled by a 2D parallel plate model. The analysis performed in Ref. [6] shows that for convective levels corresponding to our calculations, the 2D and 3D modeling revealed no significant difference in the solute distribution, so a 2D model has been used in our simulation. From the numerical modeling, a small increase of the convective intensity, because of the solutal effect, is found, which can be generally neglected. On the ground, experiments and numerical simulation [3,5] show a strong damping solutal effect on the convection, which leads to a significant increase of radial segregation and interface curvature during the solidification of concentrated $Ga_{1-x}In_xSb$ ($x = 0.1$ and $x = 0.2$) alloys. On the contrary, in space, because the configuration is rather horizontal Bridgman, the level of the convection is almost constant during the solidification, so the increase of the radial segregation and interface curvature are limited [5].

In this paper, the numerical investigations are extended by introducing the analysis of the g-jitter effect on the chemical segregation. This study is related to experimental observations, which show interesting segregations phenomena, 90 min after the growth process, when the microgravity measurements versus frequency show a sharp increase of the g-jitter amplitude.

2. Physical and numerical model

The numerical simulation is performed by using the commercial code FIDAP, for a two-dimensional domain which contains the sample and the crucible. Because the study is focused on the segregation phenomena, observed at the beginning of the solidification, the numerical analysis is restricted to the boron nitride part of the crucible. The experimental growth parameters corresponding to the numerical modeling are

- pulling rate $V = 1.1\,\mu m/s$;
- thermal axial gradient in the adiabatic zone $G_T = 60\,K/cm$;
- sample diameter 1.2 cm;
- sample length 6 cm including 1.5 cm of seed;
- BN crucible thickness 0.2 cm.

The physical properties of $Ga_{1-x}In_xSb$ pseudo-binary alloys and BN crucible are given in Refs. [3,7].

The equations describing the heat, momentum and solute transport during the solidification are

$$\rho c_P\left(\frac{\partial T}{\partial t} + \vec{u}\nabla T\right) = k\nabla^2 T, \tag{1}$$

$$\rho\left[\frac{\partial \vec{u}}{\partial t} + (\vec{u}\nabla)\vec{u}\right] = -\nabla p + \mu\nabla^2\vec{u} + \rho\vec{g}[1 - \beta_T(T - T_0) + \beta_C(C - C_0)], \tag{2}$$

$$\frac{\partial C}{\partial t} + \vec{u}\nabla C = D\nabla^2 C, \tag{3}$$

where ρ, k, c_P, β_T, β_C, μ, and D are the density, thermal conductivity, specific heat, thermal expansion coefficient, solutal expansion coefficient, viscosity and diffusion coefficient, respectively.

The experimental data for the temperature evolution around the crucible measured by the thermocouples during the solidification are imposed as boundary conditions on the crucible external wall. The no-slip and, no-flux conditions for velocity and solute field, respectively, are imposed on the solid boundaries of the liquid domain. In practice, dewetting occurred [1], however, leading to only 50 μm of free surface, which is neglected in this analysis. At the solid–liquid interface, the thermal flux balance and solute balance conditions are

$$(k_S \nabla T|_S - k_L \nabla T|_L)\vec{n} = \rho_S \cdot \Delta H \cdot (\vec{V} \cdot \vec{n}), \quad (4)$$

$$D \nabla C|_L \vec{n} = (1 - K)(\vec{V} \cdot \vec{n}) C_I. \quad (5)$$

The solute effect on the melt convection is described by the density term $\rho\beta_C(C - C_0)$ in the momentum equation. The variation of the melting point as function of the interface concentration is also taken into account by considering a linear dependence

$$T_S(C) = T_S^0 - m_L C_I. \quad (6)$$

T_S^0 is the melting point of the pure material (GaSb) and m_L is the slope of the liquidus curve in the GaInSb phase diagram [8].

Numerical transient modeling is performed by using the commercial software FIDAP. The governing Eqs. (1–3) written in the non-dimensional form are solved by using the finite element method. The mesh with 6000 quadrilateral elements has the density of nodes improved in the region of the interface. For the transient modeling which supposes a moving boundary, e.g. solid–liquid interface, the FIDAP deformed mesh technique is used.

3. Results and discussion

The analysis is performed in two steps. In the first part, the effect of microgravity variations on the thermosolutal convection and chemical segregations is studied in our configuration. In the second step, the analysis of the specific experimental chemical behavior is performed.

3.1. Effect of g-jitter

In this first paragraph, the general effect of variation of gravity level and frequency in the ESA8 experiment is studied. These general gravity levels and frequencies are given in Ref. [9]. The analysis of microgravity residual accelerations performed in Ref. [9], shows that the microgravity vector g at a certain location within the spacecraft can be written as

$$g = \sum (g_S + g_T), \quad (7)$$

where g_S are the quasi-steady accelerations with frequencies lower than the orbital frequency ($f_S = 2 \times 10^{-4}$ Hz) and g_T are the transient accelerations, called the g-jitter, which are characterized by a broad frequency spectrum. The microgravity measurements performed onboard the Spacelab during the STS78/LMS mission, show an average steady value of gravity vector $g_S \approx 10^{-6} g_0$ at the frequencies $f < f_S$. In the following, the reference quasi-steady case is simulated by applying a gravity vector $g_{q.s.}$ ($10^{-6} g_0$, 10^{-4} Hz).

The numerical investigations performed in Ref. [10] related to the effect of single and multi-component oscillatory disturbances on the melt convection and solute segregation, show a small sensitivity of the Bridgman model to the multiple frequency oscillatory disturbances. Therefore, only single frequency disturbances have been taken into account in our simulation.

The g-jitters effect is simulated by considering a sinusoidal variation versus time of both x and y-axis components of the gravity vector

$$g_x = g_A \sin 2\pi f \cdot t, \quad (8)$$

$$g_y = g_A \sin 2\pi f \cdot t, \quad (9)$$

The analysis is performed by varying the amplitude of the gravity vector between 10^{-6} and $10^{-2} g_0$ and the frequency between 10^{-4} and 10 Hz, in agreement with the experimental gravity level diagram given in Ref. [9], which allows larger accelerations at higher frequencies.

Fig. 2 shows the effect of $g(10^{-5} g_0, 10^{-2}$ Hz) periodic accelerations on the velocity field in the melt. The fluid convection is reversed at each half period. The maximum value of the convective

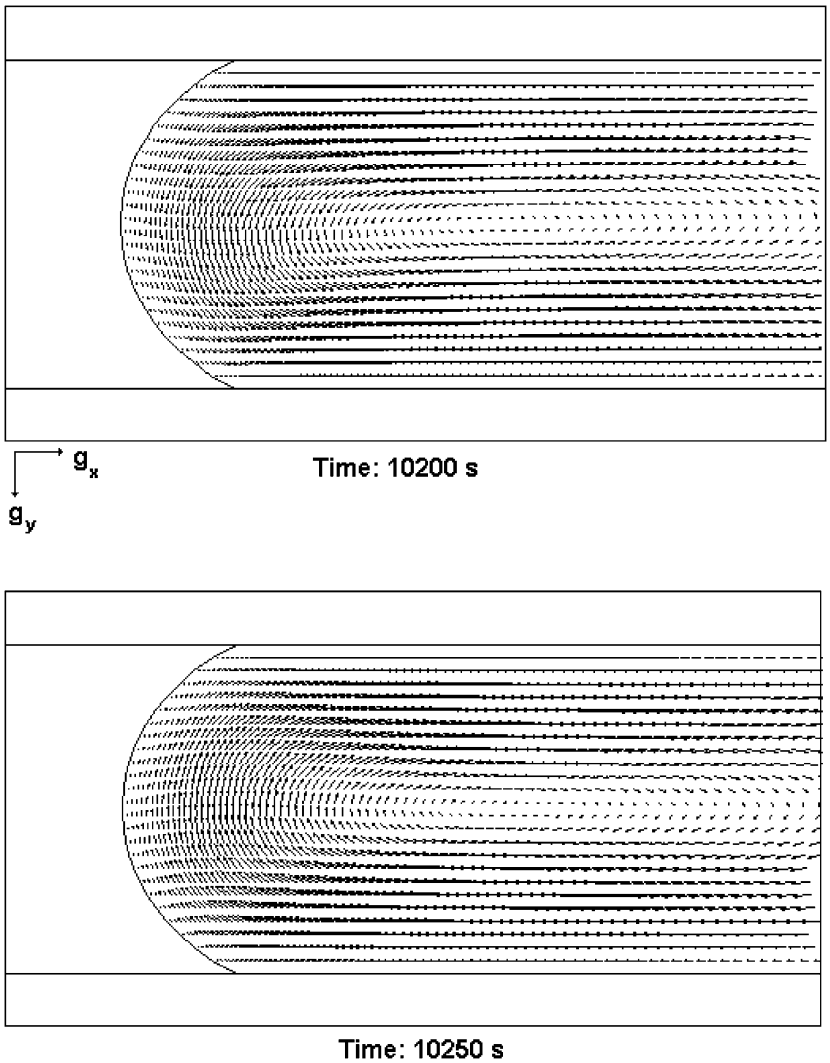

Fig. 2. Velocity field in the melt at $g(10^{-5}g_0, 10^{-2}\,\mathrm{Hz})$. Maximum convective velocity reached each half period: $u_{\max} \approx 2.3 \times 10^{-6}\,\mathrm{m/s}$.

velocity reached during each half period is $u_{\max} = 2.3 \times 10^{-6}\,\mathrm{m/s}$.

The effect of the residual gravity on the melt convection intensity and solute segregation is presented in Fig. 3. The numerical analysis is performed for the reference quasi-steady accelerations: $g_{\mathrm{q.s.}}$ and for transient accelerations: $g(10^{-5}g_0, 10^{-2}\,\mathrm{Hz})$, $g(10^{-4}g_0, 1\,\mathrm{Hz})$ and $g(10^{-2}g_0, 10\,\mathrm{Hz})$. In Fig. 3a, the maximum value of the convective velocity $u_{\max}$ is plotted versus g. It can be observed that a 10-time increase of $u_{\max}$ in the case of transition between quasi-steady acceleration to $g(10^{-5}g_0, 10^{-2}\,\mathrm{Hz})$. In the case of stronger accelerations at higher frequencies, the increase and the convective intensity is not important (only three times) and the maximum convective velocity has the same order of magnitude as for $g(10^{-5}g_0, 10^{-2}\,\mathrm{Hz})$: $u_{\max} \approx 10^{-6}\,\mathrm{m/s}$. At the same amplitude of the gravity vector $g_A = 10^{-5}g_0$, the increase of the frequency from $10^{-2}\,\mathrm{Hz}$ to $10^{-1}\,\mathrm{Hz}$, leads to a decrease of the convective level. So, the effect of higher frequency residual acceleration on the melt

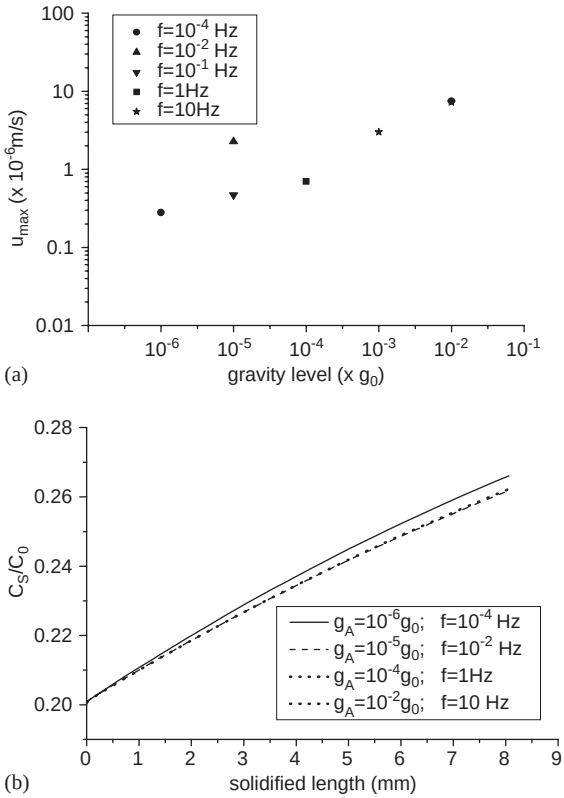

Fig. 3. Effect of the residual gravity on the melt convection and solute segregation: (a) maximum convective velocity as function of the gravity amplitude for frequencies between 10^{-4} Hz and 10 Hz; (b) axial profile of solute in the solid sample for different gravity levels.

convection is small, as already reported for purely thermal convection in Ref. [10].

The same conclusion is obtained from the analysis of the axial and radial segregations computed by changing the gravity level. As can be seen in Fig. 3b, the most noticeable difference in the axial profile of solute in the crystal is observed when the gravity level is changing from $g_{\text{q.s.}}$ to $g(10^{-5}g_0, 10^{-2}\,\text{Hz})$. In this case, the increase of the convection intensity leads to a small decrease of the axial segregation. The radial segregation, defined as $\delta C = (C_{\max} - C_{\min})/C_{\text{av}}$, where $C_{\max}$, $C_{\min}$ and C_{av} are the maximum, minimum and the average concentration in the melt at the interfaces, respectively, is also reduced from $\delta C = 0.14$ at $g_{\text{q.s.}}$ to $\delta C = 0.12$ at $g(10^{-5}g_0, 10^{-2}\,\text{Hz})$. For higher frequencies $f > 10^{-2}$ Hz, the effect of the transient accelerations is very small and the axial solute profile is practically unchanged. This result is in agreement with the cutting frequency $f_C \approx 6 \times 10^{-3}$ Hz, calculated from the formulas proposed in Ref. [11]. From these results, it appears that the effect of g-jitters on the solutally and thermally driven convection and related chemical segregations in our experiment can be, in a first approximation, simulated by a single transient acceleration, $g(10^{-5}g_0, 10^{-2}\,\text{Hz})$, which has practically the same effect on the flow and solute distribution as all the g-jitter spectrum analyzed in Fig. 3.

3.2. Analysis of segregation phenomena during flight experiment

The chemical characterization of the flight sample shows a complicated behavior of the axial solute distribution at the beginning of the solidification (see Fig. 4). When the growth process is started, a significant increase of the axial In composition is observed on the first 3 mm solidified length. After that, the solute profile shows a destabilized growth on the following 4 mm, then a decrease of the axial In concentration which indicates an intense solute mixing in the melt. After the chemical homogenization of the liquid sample, the stable growth process, with a planar front, is restarted with an axial profile which corresponds to Scheil's law

$$C_S = KC_0(1 - f_S)^{1-K}, \tag{10}$$

where f_S is the solidified fraction and C_0 is the initial concentration in the liquid sample.

These experimental observations are studied by using the transient numerical analysis of the coupled thermal, hydrodynamical and solute segregation phenomena.

The significant increase of the axial In concentration after the start of the solidification cannot be explained as a result of the residual accelerations induced flow, even at the low level of the gravity measured at this time ($g \approx 10^{-6}g_0$). This profile can be explained as an effect of the partial chemical mixing between the melted seed (8% initial solute concentration) and the liquid feed

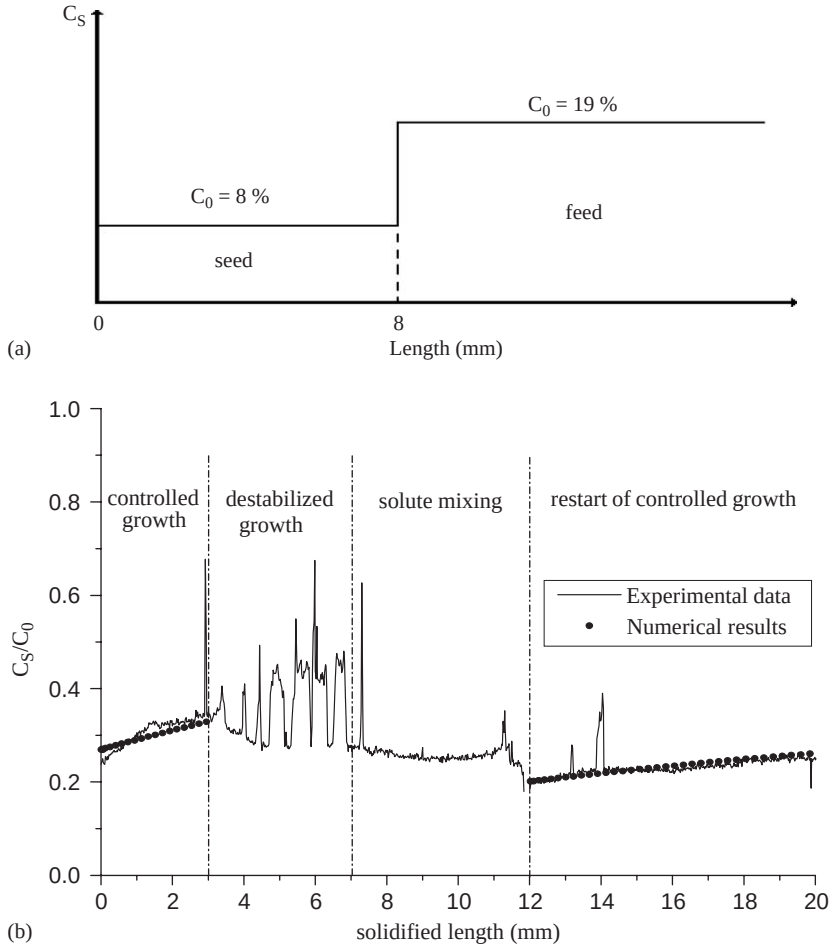

Fig. 4. (a) Initial composition profile before the seed melting. (b) Experimental and numerical computed longitudinal profile of In in the solid sample at the beginning of the solidification. The gravity level increases 90 min after the start of the growth process.

(19% initial solute concentration), because of insufficient homogenization time after the remelting. From the numerical modeling, it is found that the liquid–solute concentration is not homogeneous at the start of growth and varies near the interface; a liquid layer having an ascendant In profile rises in this region. When the growth process is started, the liquid layer is solidified with a quick increase of solute concentration as a result (see experimental and numerical results in Fig. 4b).

The destabilized growth begins 90 min after the growth process start, when the microgravity measurements versus frequency and time show almost 10-time increase of g-jitter amplitude, as the probable result of the crew activities' intensification [4]. In agreement with the numerical investigations performed in the previous paragraph, this increase of gravity level has been modeled as a rapid transition from the quasi-steady accelerations $q_{\mathrm{q.s.}}$ to periodic accelerations $g(10^{-5}g_0,\ 10^{-2}\,\mathrm{Hz})$. The numerical simulation shows a significant increase of the convective level at this moment, which leads to a rapid decrease of the solute concentration in the melt near the interface. Therefore, because of the melting point dependence on the solute concentration (see Eq. (6)), the solid–liquid interface jumps forward. A constitutional supercooling with a morphological destabilization of the interface may therefore occur. This phenomenon is indeed

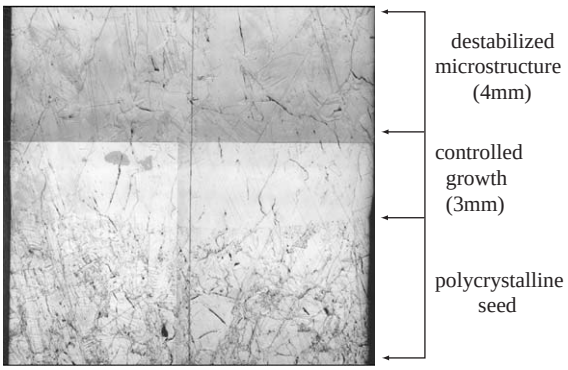

Fig. 5. Photograph of the beginning of the solid sample after cutting, polishing and etching.

observed from the longitudinal profile of In concentration (see Fig. 4) and also from the metallographic analysis of the etched sample, which shows a destabilized structure after about 3 mm of crystal length (see Fig. 5).

After this destabilized growth, the solidification with a planar front is restarted and the still active intense mixing of the solute leads to a decrease of the axial concentration on about 5 mm length of further growth, in agreement with the characteristic mixing time given by the simulation. After the homogenization of the solute in the liquid sample, the controlled growth with a characteristic Scheil's profile of the solute occurs till the end of the solidification. The measured axial concentration in the crystal, indicates an average composition of the melt $C_0 \approx 10\%$, after the complete homogenization of the liquid sample.

3.3. Analysis of morphological destabilization

In order to better study the occurrence of the morphological destabilization phenomenon caused by the rapid increase of the convective intensity, a numerical simulation is performed by changing the gravity amplitude. According to the stability criterion [12], the solid–liquid interface is morphologically stable when

$$C_I V_I < \frac{D G_T}{m_L (1-K)}, \tag{11}$$

where C_I is the solute concentration in the liquid at the interface and V_I is the interface velocity.

So, a critical value $(C_I V_I)^{cr} = D G_T / m_L (1-K)$ can be computed for our configuration by using the value of G_T carried out from the numerical computation of the thermal field.

In Fig. 6, the time evolution of the product $C_I V_I$ is plotted before and after the moment when the gravity level is changed ($t = 90$ min). As shown in this figure, before $t = 90$ min, a small increase of the $C_I V_I$ product is observed due to the solute segregation at the interface. At $t = 90$ min, the amplitude of the transient accelerations is suddenly varied.

When the acceleration is $g_A = 2 \times 10^{-6} g_0$, the increase of the convective level leads to a small decrease of the solute concentration near the interface. In this case, the interface velocity is also reduced, so the quantity $C_I V_I$ is below the critical value and the interface remains morphologically stable. If g_A is $5 \times 10^{-6} g_0$, the interface velocity increases, but $C_I V_I$ quantity is still less than $(C_I V_I)^{cr}$. At a gravity level $10^{-5} g_0$, the level of the convection increases by 10 times and a significant increase of V_I is obtained from the simulation. In this case $C_I V_I > (C_I V_I)^{cr}$, so a morphological destabilization of the interface may occur. Then the liquid concentration at the interface decreases quickly and the interface

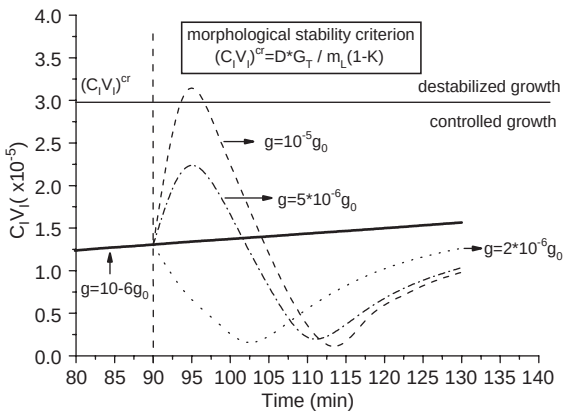

Fig. 6. Numerical analysis of the morphological destabilization phenomena caused by the rapid increase of the convection. The amplitude of the residual gravity imposed after $t = 90$ min is varied between 2×10^{-6} and $10^{-5} g_0$.

supercooling as well, by 10 K, which corresponds, taking into account the thermal gradient, to 4 mm of the destabilized growth as seen in Fig. 4b.

In conclusion, the stability of the interface is related to variations in the convective level and a rapid and significant increase of the flow intensity can produce a morphological destabilization of the interface during the solidification of concentrated alloys. This effect can occur during the growth process, when a diffusive layer is already built up near the interface at the moment when the increase of the convective level is produced.

4. Conclusions

Numerical simulation has been used in order to investigate the segregation phenomena observed during the space experiment on the Bridgman solidification of $Ga_{0.9}In_{0.1}Sb$ concentrated alloy.

The analysis of the melt flow induced by the residual gravity, shows a significant increase (10 times) of the flow velocity when a transition from quasi-steady accelerations $g(10^{-6}g_0,\ 10^{-4}\,\mathrm{Hz})$ to the periodic accelerations $g(10^{-5}g_0,\ 10^{-2}\,\mathrm{Hz})$ occurs. A further increase of the g-jitters amplitude and frequency has no significant influence on the convective level. From the numerical simulation, it is found that only a 3-time increase of the convective intensity for $g(10^{-2}g_0,\ 10\,\mathrm{Hz})$ takes place as compared with $g(10^{-5}g_0,\ 10^{-2}\,\mathrm{Hz})$ gravity level. However, this increase of the convection has a partially undetectable influence on the axial and radial solute distribution in the crystal. The numerical computation, in agreement with experimental results, shows that an axial solutal profile described by Scheil's law for a convecto–diffusive regime can be obtained even at low levels of the gravity 10^{-6}–$10^{-5}g_0$. Only a small decrease of the axial and radial segregations is observed when the g-jitters effect is taken into account in the simulation.

However, a sharp increase of the gravity level during the solidification process of the concentrated alloys can produce interesting segregation phenomena, such as a morphological destabilization of the solid–liquid interface. This phenomenon has been experimentally observed for the first time, 90 min after the start of the growth process, when the gravity measurements indicate a sudden increase of the g-jitters accelerations. In this paper, a physical explanation of this effect has been provided from the numerical modeling, it is found that the rapid increase of the flow intensity leads to a decrease of the solute concentration near the interface and this induces a constitutional supercooling phenomenon. An analysis of this effect is performed by varying the amplitude of the gravity and it is found that a rapid and significant increase of the convection intensity can produce a morphological destabilization of the interface during concentrated alloys solidification.

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